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Job Description: Full Stack Engineering Intern (AI Agentic Platforms)

Location: Bengaluru
Duration: 6 months to 1 year (extendable / convertible), PPO will be given based on performance.

Stipend : 30k per month/-

About Tredence Studio

At Tredence Studio, we are not just following trends; we are building the future of AI. We are developing State-of-the-Art (SOTA) capabilities in **AI Agentic Engineering** to build a paradigm-shifting platform from the ground up. We are looking for an elite **Full Stack Engineering Intern** who is ready to move beyond coursework and solve complex, real-world problems. This is a zero-to-one role, and you will be a core contributor to a product built to challenge industry leaders.

The Opportunity

This is not a typical internship. You will not be on the sidelines. You will be in the trenches with a team of expert engineers, tackling difficult architectural and product challenges. You will gain deep, hands-on mastery of the entire development lifecycle, from architecting scalable frontend components in **React** and **TypeScript** to engineering high-performance **Python** backends, and deploying them on a robust **Kubernetes** infrastructure. We are seeking a candidate with a relentless drive to learn and a passion for building truly exceptional software.

Core Responsibilities

- **Architect and Develop:** Design, build, and scale critical application features across the full stack using **React, Next.js, TypeScript, Node.js,** and **Python (FastAPI)**.
- **Master Clean Architecture:** Go beyond basic coding to implement and uphold elite standards of software design, including strong typing, modular design, and creating highly maintainable and performant code.
- **Engineer Advanced UIs:** Build highly responsive, reusable, and scalable UI components that are pixel-perfect representations of our **design systems**.
- **Build Robust Backends:** Design and implement complex backend APIs, sophisticated data models, and high-throughput integration workflows. You will work extensively with **SQL (PostgreSQL)** and **NoSQL (Firestore)** databases.

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- **Implement Real-Time Systems:** Engineer and optimize real-time features using **WebSockets** and **Server-Sent Events (SSE)** to deliver instantaneous data to users.
- **Secure and Deploy:** Integrate and manage secure authentication and authorization flows (**OAuth/OIDC/JWT**) with **Azure**. You will gain hands-on experience deploying and managing services in a **cloud-native environment (Kubernetes)**.
- **Own Code Quality:** Drive code quality by writing comprehensive **unit tests (Jest/RTL)** and **E2E tests (Cypress/Playwright)**, ensuring the reliability of our platform.
- **Solve Hard Problems:** Apply your deep understanding of **Data Structures, Algorithms,** and architectural patterns to solve complex performance and scalability challenges.

Required Skills & Qualifications

- Pursuing or recently completed a B.S., M.S., or Ph.D. in Computer Science or a related quantitative field.
- Demonstrable, high-level proficiency in **React, TypeScript,** and **Python**, showcased through a strong portfolio of projects (e.g., a detailed GitHub profile).
- Deep understanding of CS fundamentals, including **Data Structures, Algorithms, operating systems, and computer networks**.
- Strong foundation in modern web technologies, including **HTML, CSS (Tailwind),** and responsive design principles.
- A provable passion for building products from the ground up and a "zero-to-one" mindset.
- Excellent problem-solving abilities, with a talent for structured, analytical debugging.

Preferred (Bonus) Qualifications

- Experience contributing to **open-source projects**.
- Hands-on experience with **Kubernetes, Docker,** and building/managing **CI/CD pipelines** (e.g., GitHub Actions, Jenkins).
- Familiarity with advanced architectural patterns like **monorepos (Nx/Turborepo)** or **micro-frontends**.
- Knowledge of advanced optimization techniques, including **Server-Side Rendering (SSR)/Static Site Generation (SSG)** and scalable API design.
- Exposure to building and maintaining design systems with tools like **Storybook**.
- Experience with cloud monitoring, logging, and alerting tools.

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Case Study – Full stack Engineering Intern

Build Prototype: HR Workflow Designer Module (React + React Flow)

Objective

Design and implement a mini–HR Workflow Designer module where an HR admin can visually create and test internal workflows such as onboarding, leave approval, or document verification.

You will build a functional prototype that demonstrates:

- Deep knowledge of React and React Flow
- Ability to architect modular, scalable front-end systems
- Ability to integrate mock APIs
- Ability to build configurable nodes with custom edit forms
- Ability to simulate/test designed workflows in a small sandbox

This exercise is time-boxed to 4–6 hours. Focus on architectural clarity and working functionality, not pixel-perfect UI.

Deliverables

A GitHub repository (or zipped project) that includes:

1. React application (Vite or Next.js preferred)
2. React Flow canvas with multiple custom nodes
3. Node configuration/editing forms for each node type
4. Mock API integration
5. Workflow Test/Sandbox panel
6. README explaining architecture, design choices, and assumptions

No authentication or backend persistence is required.

Functional Requirements

1. Workflow Canvas (React Flow)

Implement a drag-and-drop workflow canvas with:

Node Types

- Start Node – workflow entry point
- Task Node – human task (e.g., collect documents)
- Approval Node – manager or HR approval step

- Automated Step Node – system-triggered actions (e.g., send email, generate PDF)
- End Node – workflow completion

Supported Canvas Actions

- Drag nodes from a sidebar onto the canvas
- Connect nodes with edges
- Select a node to edit it
- Delete nodes/edges
- Auto-validate basic constraints (e.g., Start Node must be first)

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2. Node Editing / Configuration Forms (Key Requirement)

Each node type must have an editable “Node Form Panel” that appears when the node is selected.

Minimum Required Fields

Start Node

- Start title
- Optional metadata key-value pairs

Task Node

- Title (required)
- Description
- Assignee (string input)
- Due date (text input or simple date field)
- Optional custom fields (key-value)

Approval Node

- Title
- Approver role (e.g., “Manager”, “HRBP”, “Director”)
- Auto-approve threshold (number)

Automated Step Node

- Title
- Choose an action from a mock API list
- Action parameters (dynamic based on mock action definition)

End Node

- End message

- Summary flag (boolean toggle)

Evaluation: dynamic forms, controlled components, clean state handling, type safety, modularity.

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3. Mock API Layer

Create a lightweight API layer using JSON server, MSW, or local mocks.

Must support:

- GET /automations
  Returns mock automated actions such as:
- [
- { "id": "send_email", "label": "Send Email", "params": ["to", "subject"] },
- { "id": "generate_doc", "label": "Generate Document", "params": ["template", "recipient"] }
-]
- POST /simulate
  Accepts workflow JSON and returns a mock step-by-step execution result.

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4. Workflow Testing / Sandbox Panel

A small panel or modal that:

- Serializes the entire workflow graph
- Sends it to the mock /simulate API
- Displays a step-by-step execution log (basic text or timeline UI)
- Validates structure (e.g., missing connections, cycles)

This tests the candidate's reasoning around state, graph structures, and API workflows.

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5. Architecture Expectations

We will evaluate:

- A clean folder structure
- Separation of canvas logic, node logic, and API logic
- Reusable custom hooks
- Ability to design abstractions that scale
- Thoughtful component decomposition
- Form structure that can be extended to new node types

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- Clarity of interfaces/types for workflow nodes

Optional (Bonus, Not Required)

If time permits:

- Export/Import workflow as JSON
- Node templates
- Undo/Redo
- Mini-map or zoom controls
- Workflow validation errors visually shown on nodes
- Auto-layout
- Node version history

Assessment Criteria

Area	What We Look For
React Flow proficiency	Custom nodes, positioning, edge management
React architecture	Hooks, context, components, folder structure
Complex form handling	Node forms, dynamic fields, validation
Mock API interaction	Data layer, async patterns, abstraction
Scalability	Is the solution extensible?
Communication	README, assumptions, design notes
Delivery speed	Can a senior engineer ship value fast?

What the Candidate Submits

- GitHub repo or zipped project
- README explaining:
    - Architecture
    - How to run
    - Design decisions
    - What they completed vs. what they would add with more time

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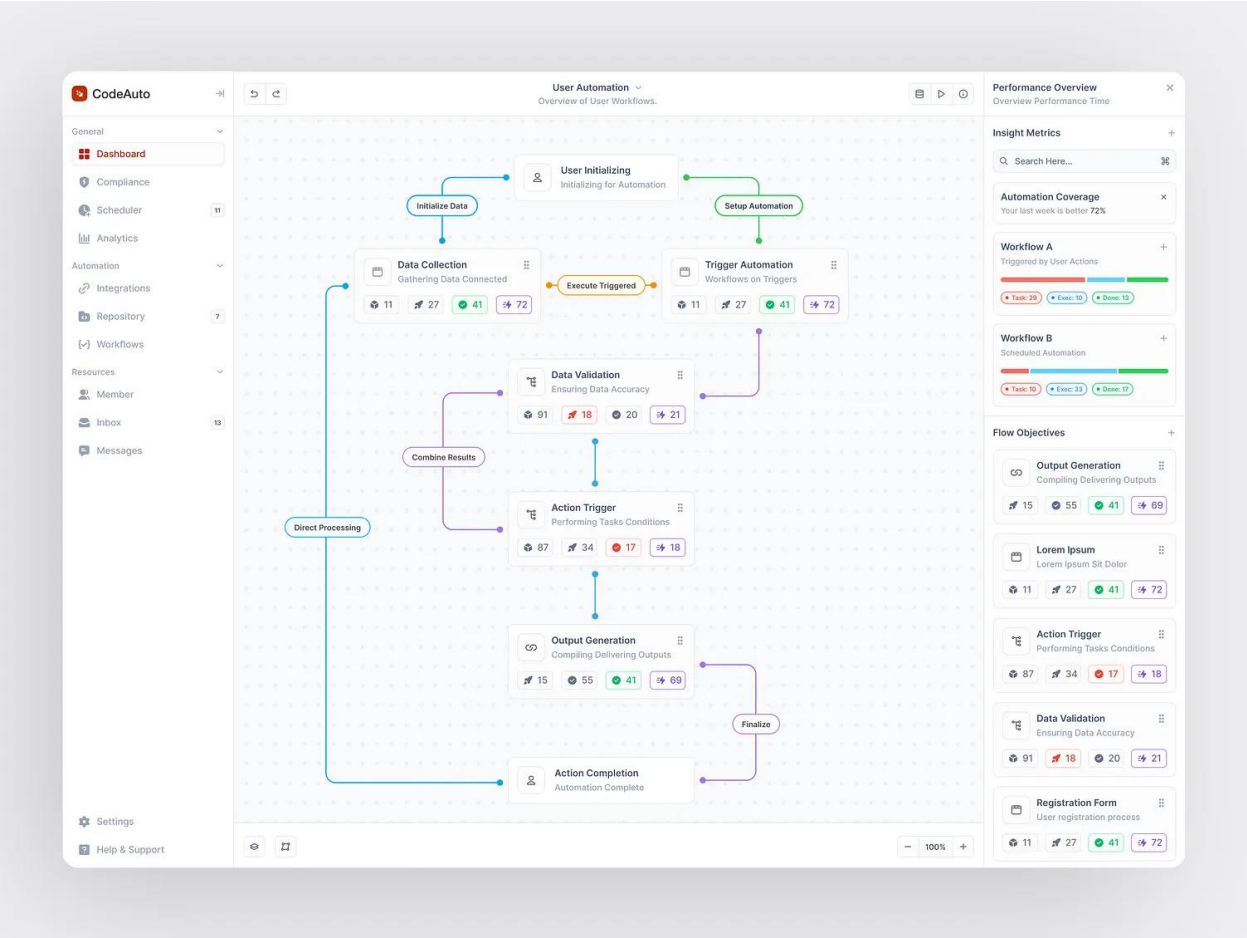
UI Reference :

Reference 1

- ***Use this <u>as a broad reference</u> for the UI, Theme and the Canvas. This is not the design spec***
- Your solution need not implement all the Specs implemented in this UI and this is only to be used as design reference.
- Improvements and innovations on this design reference is encouraged and incentivized



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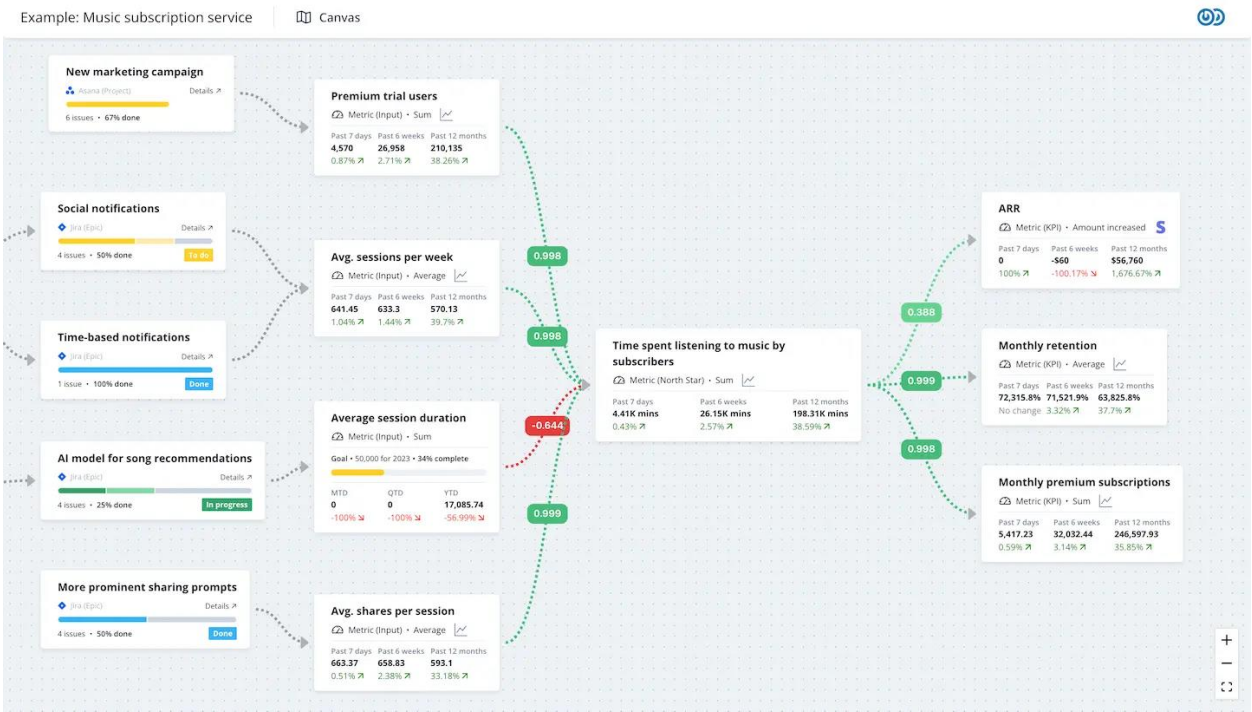
Reference 2



Instructions :

Dear Student,

Tredence Studio's AI Agents Engineering team is inviting applications for the **AI Engineering Internship – 2025 Cohort**.
This internship offers a rare opportunity to work on **frontier engineering and research** in the fast-growing domain of **AI Agent Engineering**.

If you are genuinely passionate about building the next generation of intelligent systems, this is for you.

We're opening a Frontend Internship Project for students who genuinely enjoy building web apps and APIs and have spent a few years building things in React/TypeScript on their own — college projects, freelance work, side apps, anything real.

This is a practical, hands-on internship. You'll get to build real UI modules for our AI products and work closely with senior engineers.

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Who this is for :

- Students who have been coding frontend apps for 3–5 years
- People who can take a small requirement and turn it into a working UI
- Students who like debugging, polishing components, and writing clean code
- Basic understanding of APIs, SQL etc.
- Strong DSA and Problem solving skills

Bonus (Not mandatory) :

- Experience in writing REST APIs in Python, NodeJs
- Experience in Rust

Nice to have: Tailwind, Zustand/Redux, Vite, and some UI/UX sense.

To Apply, Reply with:

1. GitHub link to your project
2. Any other apps or repos you've built
3. A short note on one tricky frontend bug you solved
4. Your resume or LinkedIn

We're looking for students who have genuinely spent time building things and want to work on real features. If that's you, would love to see your submission.

Note : Kindly submit your GitHub URLs without fail

If you believe you have the depth, passion, and discipline to work on cutting-edge AI Agent systems, we'd love to see your application.

Best regards,
Tredence Studio — AI Agents Engineering Team